seed42 Audio Module

COMP3850 PACE, Groups 21

We now all work on the `main` branch, with no feature branches. Everyone pulls the latest code, does their task, and pushes back to `main`. Follow the setup steps once, then use the git routine every time you work. Read `AGENTS.md` before writing any code.

1 Get the code

Open the Anaconda Prompt, move into your project folder, and pull the latest from GitHub. Do this at the start of every session so you have everyone's newest work.

```
cd Documents\Project\seed42
git pull
```

Please note that this is taken from my folders, not yours, so make sure to adjust it

2 Install Anaconda and the packages

Download and install Anaconda from `anaconda.com`. To keep things simple, we install the packages into the Anaconda **base** environment, so there is no separate virtual environment to manage.

Open the **Anaconda Prompt**, move into the project folder, and run these once:

```
cd Documents\Project\seed42

pip install librosa numpy scipy soundfile pyyaml audioread
conda install -c conda-forge ffmpeg
pip install -e .
```

What each line does:

- `pip install librosa ...` installs the audio libraries we use.
- `conda install ... ffmpeg` adds mp3 decoding support on Windows.
- `pip install audioread` lets librosa reach ffmpeg for mp3 files.
- `pip install -e .` installs our own `seed42_audio` package so that `import seed42_audio` works from anywhere. This needs `pyproject.toml` in the project root, which is already in the repo.

Note. For testing, a short `.wav` file is the most reliable on Windows. If mp3 still fails after installing ffmpeg, use a `.wav` instead.

3 Set up the Interactive Window in VS Code

We run code with Shift and Enter in the Interactive Window, not in the terminal. Set this up once:

1. In VS Code, install the **Python** and **Jupyter** extensions (Extensions panel, search each, Install).
2. Select the interpreter: press Ctrl+Shift+P, type `Python: Select Interpreter`, and choose the Anaconda **base** environment (the path contains `anaconda3`).

3. Turn on Shift and Enter to the Interactive Window: open Settings (Ctrl+comma), search `execute selection`, and enable `Jupyter: Interactive Window` > `Text Editor: Execute Selection`.

Now, with a `.py` file open, Shift and Enter runs the selected code in the Interactive Window.

Note on running. Files that end with `if __name__ == "__main__":` only run their body when launched as a script. To test in the Interactive Window, run the lines directly without that guard.

4 Read AGENTS.md before coding

Before you write or change any code, open `AGENTS.md` in the project root and follow it. The key rules: use only the past few seconds of audio (causal), stay on the Phase 1 trained-model work, keep code simple, and use no em dashes anywhere. If you use an AI assistant, tell it to read `AGENTS.md` first.

5 The git routine (every time)

Because we all share `main`, always pull before you push, or your push may be rejected. In the Anaconda Prompt, from the project folder:

```
cd Documents\Project\seed42

git pull                        # get everyone's latest first
git status                      # see what you changed
git add .                       # stage your changes
git commit -m "add your comment"
git pull                        # pull again in case someone just pushed
git push                        # send your work to main
```

If the push is rejected with a message about updates being rejected, run `git pull` then `git push` again. If git reports a conflict (the same lines changed by two people), stop and ask Jayden or Hatim rather than guessing.

Note. `git status` should not list your `.venv`, audio files, or cache folders; those are ignored on purpose. Only commit your own work.

6 Run it and confirm

The library files `io/loader.py` and `io/stream.py` are imported, not run directly. The file you run is `scripts/run_stream.py`. In the Interactive Window, run these lines (make sure the filename matches the audio file in your `data` folder):

```
from seed42_audio.io.stream import AudioStream

stream = AudioStream("data/test.wav", window=6.0, hop=1.0)
for timestamp, segment in stream.segments():
    print(f"t={timestamp:5.1f}s   {len(segment)} samples")
```

Success looks like a rolling list:

```
t=  6.0s   132300 samples
t=  7.0s   132300 samples
t=  8.0s   132300 samples
```

Take a screenshot of this output and post it in the group chat to confirm your setup works.

If you get an error: check the file location first

Notebooks and the Interactive Window do not always start in the project root, so a relative path like `data/test.wav` can fail to find the file. Before assuming the code is broken, run this to see where you are and whether the file is found:

```
import os
print(os.getcwd())
print(os.path.exists("data/test.mp3"))
print(os.listdir("data"))
```

If `os.getcwd()` does not end in `seed42`, or `os.path.exists` prints `False`, you are in the wrong folder. Move to the project root, then run again:

```
import os
os.chdir(r"C:\Users\yourname\Documents\Project\seed42")
print(os.getcwd())
```

Do not commit that `os.chdir` line into shared code, since the path is specific to your machine.